

Notes: Hubbard (QJE, 1986)

Supply Shocks and Price Adjustment in the World Oil Market

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1 Big picture

- **Question.** How do transitory oil supply shocks affect world oil prices when the oil market has two coexisting prices: a short-term spot price and a longer-term contract price?
- **Setting.** The paper studies the world oil market in the 1970s and early 1980s, when most crude oil transactions occurred through term contracts, while a smaller spot market absorbed short-run imbalances.
- **Core puzzle.** A supply disruption is temporary, so a frictionless one-price market would imply a sharp but temporary price movement. Yet the observed oil market showed that shocks could have persistent effects on official or contract prices. The paper asks how this persistence can arise from the market's pricing structure.
- **Main mechanism.** Contract prices do not adjust one-for-one and immediately to every spot-market fluctuation. They are updated using past prices, expected future prices, and information about market imbalance. This gradual adjustment means even white-noise shocks can have effects that persist in long-term prices.
- **Key distinction.** The spot price clears the marginal spot market, but it is not necessarily the relevant marginal cost for all buyers. A buyer can obtain oil through a contract only by committing to a longer-term arrangement. Therefore the economically relevant price is a weighted average of spot and contract prices.
- **Central result.** A negative supply shock makes the spot price jump immediately, but the contract price adjusts more smoothly and persistently. As the contract price rises, it changes later spot-market clearing conditions. Hence spot and contract prices are jointly determined, not a simple recursive system in which the spot price mechanically drives the contract price.
- **Comparative statics.** Persistence is larger when contract sellers put more weight on past prices, when a larger share of trade is done through contracts, and when oil demand is less price-responsive. Persistence is smaller when the spot market becomes more important or when demand is more price elastic.
- **Policy relevance.** The effect of tariffs, public stockpile releases, or other emergency policies depends on the contracting structure of the market. A policy trigger based only on the spot price can be misleading if the share of spot trade changes over time.
- **Empirical message.** Simple regressions of contract prices on lagged spot prices are likely unstable across market regimes. The paper illustrates this with monthly oil price data from 1974–1984: coefficients change substantially before and after 1980.
- **Takeaway.** The paper is a model of price adjustment under partial price rigidity. Oil shocks affect prices not only through contemporaneous scarcity but also through the institutional structure of trades.

2 Data and measurement (key objects)

2.1 Paper type and empirical scope

This is primarily a **theoretical paper** with an illustrative empirical section. It does not estimate a full structural model of the world oil market. Instead, it builds a simple dynamic model of a two-price commodity market and then uses oil price data to show why reduced-form price regressions can be unstable.

The empirical material has two main roles:

- to document that spot and contract oil prices often diverged during shocks;
- to show that the statistical relation between lagged spot prices and contract prices changed across subperiods.

2.2 Main price objects

There are two price concepts.

Spot price, P_t^S . This is the price for single-cargo or very short-term transactions. In the historical oil market, spot transactions were used largely to balance short-term shortages and surpluses.

Contract price, P_t^C . This is the price in ongoing term contracts. The paper often refers to official government selling prices as the empirical counterpart.

The model's important point is that the spot price is not automatically the relevant marginal price for all purchase decisions. Buyers choosing between spot and contract transactions face a tradeoff between today's spot purchase and a longer-term commitment.

2.3 Composite price

The economically relevant price for demand is a weighted average:

$$P_t = \alpha P_t^C + (1 - \alpha) P_t^S, \quad (1)$$

where α is the fraction of world oil trade executed through long-term contracts.

Interpretation.

- If α is high, most trade is under contracts and the spot market is small.
- If α is low, spot transactions are more important and prices are more flexible.
- Changes in α change both the size of spot-price jumps and the persistence of shocks.

2.4 Shock variables

The model includes two transitory shocks:

$$\varepsilon_t^D \quad \text{and} \quad \varepsilon_t^S. \quad (2)$$

Here ε_t^D is a demand shock and ε_t^S is a supply shock. They are assumed to be independently and identically distributed, mean zero, and not serially correlated.

A negative oil supply disruption corresponds to:

$$\varepsilon_t^S < 0. \quad (3)$$

Interpretation. The paper deliberately begins with white-noise shocks. If even a white-noise disruption creates persistence in contract prices, then persistence is not coming from serial correlation in the physical shock. It is coming from the pricing institution.

2.5 Demand and spot-supply primitives

The paper uses linear demand and spot-supply responses:

$$Q^D(P_t) = A - fP_t, \quad (4)$$

$$Q^S(P_t^S) = gP_t^S. \quad (5)$$

The parameter f is the price responsiveness of demand, and g is the price responsiveness of spot supply.

Economic meaning.

- Higher f means oil demand is more elastic.
- Higher g means the spot market can supply more oil in response to a spot-price increase.
- Low f and low g make the spot price more sensitive to shocks.

2.6 Historical price data

Table I in the paper reports quarterly spot and contract prices for Mideast Light-34 crude oil from 1973 to 1984, using data from *Petroleum Intelligence Weekly*. The table highlights the following patterns:

- Spot and contract prices were close in normal periods.
- During the 1973–1974 and 1979–1980 disruptions, spot prices rose much more sharply than contract prices.
- In the early 1980s, spot prices were sometimes below contract prices, showing that the wedge can go in either direction.

2.7 Monthly regression data

Section III uses monthly data from 1974 to 1984 for Mideast Light-34 crude oil. The empirical exercise regresses the contract price on its own lag and the lagged spot price:

$$P_t^C = a_0 + a_1 P_{t-1}^C + a_2 P_{t-1}^S + e_t. \quad (6)$$

This regression is not presented as a structural model. It is used as an illustration of instability in simple econometric relationships between spot and contract prices.

3 Models and empirical specifications (all equations + interpretation)

3.1 Market-clearing condition in the spot market

Only the noncontract portion of demand must be cleared in the spot market. The spot-market clearing equation is:

$$(1 - \alpha)Q^D(P_t) + \varepsilon_t^D = (1 - \alpha)\bar{Q} + Q^S(P_t^S) + \varepsilon_t^S, \quad (7)$$

where $\bar{Q}$ is the baseline quantity associated with the initial equilibrium.

Substituting the demand and spot-supply equations gives:

$$(1 - \alpha) \left[A - f \left(\alpha P_t^C + (1 - \alpha) P_t^S \right) \right] + \varepsilon_t^D = (1 - \alpha)\bar{Q} + g P_t^S + \varepsilon_t^S. \quad (8)$$

Interpretation. The spot market clears not the entire world market, but the residual market after contracted trades are accounted for. A small spot market must absorb shocks that are large relative to its size.

3.2 Spot-price solution

Define

$$\lambda = \left[g + f(1 - \alpha)^2 \right]^{-1}. \quad (9)$$

Then the spot-price solution is:

$$P_t^S = (1 - \alpha)\lambda(A - \bar{Q}) - f\alpha(1 - \alpha)\lambda P_t^C + \lambda(\varepsilon_t^D - \varepsilon_t^S). \quad (10)$$

In deviations from the initial equilibrium:

$$p_t^S = -f\alpha(1 - \alpha)\lambda p_t^C + \lambda(\varepsilon_t^D - \varepsilon_t^S). \quad (11)$$

Interpretation.

- A positive demand shock raises the spot price.
- A negative supply shock raises the spot price because $-\varepsilon_t^S > 0$.
- A higher contract price lowers the spot price needed to clear the residual market, because the composite price facing demand is higher.

3.3 Immediate effect of a supply shock on the spot price

Differentiating the spot-price equation with respect to the supply shock gives:

$$\frac{dp_t^S}{d\varepsilon_t^S} = -\lambda - f\alpha(1 - \alpha)\lambda \frac{dp_t^C}{d\varepsilon_t^S}. \quad (12)$$

Since a negative supply shock means $\varepsilon_t^S < 0$, the spot price rises when supply is interrupted.

Why contract share matters. A higher α means a smaller spot market must absorb a larger residual imbalance. Therefore spot-price adjustment is more dramatic when more trade is locked into contracts.

3.4 Demand faced by nondisrupted producers

The paper defines the demand imbalance faced by nondisrupted producers as:

$$x_t = -fP_t + \varepsilon_t^D - \varepsilon_t^S. \quad (13)$$

Interpretation. This object summarizes market pressure. It rises when demand is unexpectedly high or when supply is unexpectedly low.

3.5 Contract-price adjustment rule

The contract price is assumed to respond to past contract prices, expected future contract prices, and demand-pressure information:

$$p_t^C = \pi_1 p_{t-1}^C + \pi_2 E_t p_{t+1}^C + \gamma (\theta_1 x_t + \theta_2 E_t x_{t+1}). \quad (14)$$

The paper imposes the simplifying restrictions:

$$\pi_1 + \pi_2 = 1, \quad \theta_1 + \theta_2 = 1. \quad (15)$$

Interpretation.

- π_1 measures how backward-looking contract pricing is.
- π_2 measures the role of expected future prices.
- γ measures how strongly contract prices respond to market-pressure information.
- θ_1 places weight on current market pressure, while θ_2 places weight on expected future pressure.

3.6 Reduced second-order difference equation

After substituting the spot-price equation into the contract-price rule, the contract-price dynamics can be summarized as:

$$Hp_t^C = \pi_1 p_{t-1}^C + BE_t p_{t+1}^C + K(\varepsilon_t^D - \varepsilon_t^S), \quad (16)$$

where

$$H = 1 + \gamma\theta_1 f\alpha [1 - f(1 - \alpha)^2 \lambda], \quad (17)$$

$$B = (1 - \pi_1) - (1 - \theta_1) f\alpha \gamma [1 - f(1 - \alpha)^2 \lambda], \quad (18)$$

$$K = \gamma\theta_1 [1 - f(1 - \alpha)\lambda]. \quad (19)$$

Interpretation. This equation is the heart of the paper. Current contract prices depend on lagged contract prices and expected future contract prices. Therefore a temporary disturbance today affects future prices through the lagged-price state variable.

3.7 Rational-expectations solution

With rational expectations and white-noise shocks, the stable solution takes the form:

$$p_t^C = \phi p_{t-1}^C + \omega(\varepsilon_t^D - \varepsilon_t^S), \quad (20)$$

where ϕ is the root inside the unit circle of:

$$B\phi^2 - H\phi + \pi_1 = 0, \quad (21)$$

and

$$\omega = \frac{K}{H - B\phi}. \quad (22)$$

Interpretation.

- ω is the immediate response of the contract price to a current market-pressure shock.
- ϕ is the persistence parameter.
- If ϕ is close to one, the contract-price effect decays slowly.
- If ϕ is close to zero, the contract-price effect dies out quickly.

3.8 Persistence of a transitory supply shock

Let $q_t = -\varepsilon_t^S$ denote the size of a negative supply shock. Then the contract-price response is:

$$\frac{\partial p_t^C}{\partial q_{t-i}} = \omega \phi^i, \quad i = 0, 1, 2, \dots \quad (23)$$

The spot-price response is:

$$\frac{\partial p_t^S}{\partial q_t} = \lambda - f\alpha(1 - \alpha)\lambda\omega, \quad (24)$$

$$\frac{\partial p_t^S}{\partial q_{t-i}} = -f\alpha(1 - \alpha)\lambda\omega\phi^i, \quad i \geq 1. \quad (25)$$

Interpretation.

- The spot price jumps when the supply shock occurs.
- The contract price rises more gradually and persistently.
- In later periods, the higher contract price reduces the spot price needed to clear the residual market.

- Thus spot and contract prices move differently after the same physical shock.

3.9 Comparative statics

The model implies three key comparative statics:

$$\frac{\partial \phi}{\partial \pi_1} > 0, \quad (26)$$

$$\frac{\partial \phi}{\partial f} < 0, \quad (27)$$

$$\frac{\partial \phi}{\partial \alpha} > 0. \quad (28)$$

Interpretation.

- More backward-looking pricing creates more persistence.
- More price-responsive demand reduces persistence.
- A larger contract share raises persistence because contracts make prices less flexible.

These comparative statics are the paper's main theoretical policy lever. Anything that changes demand responsiveness, contract duration, or the spot-contract mix can change how shocks translate into observed prices.

3.10 Econometric specification discussed in Section III

The paper discusses the reduced-form relationship:

$$P_t^C = a_0 + a_1 P_{t-1}^C + a_2 P_{t-1}^S + e_t. \quad (29)$$

This is associated with earlier work that treated lagged spot prices as driving contract prices. Hubbard's point is that this formulation is incomplete because:

- spot and contract prices are jointly determined;
- the share of spot trades, α , can change over time;
- contract adjustment parameters can change over time;
- therefore the coefficients a_1 and a_2 are not stable policy parameters.

4 Identification, interpretation, and policy logic

4.1 What kind of identification is used?

This paper is not a modern causal-identification paper. It does not use an instrument, natural experiment, or panel fixed-effects design. The key evidence is theoretical identification: the author specifies a model of how spot and contract markets clear, then derives implications for price dynamics.

The empirical section is best read as a diagnostic illustration, not as a definitive causal test.

4.2 Core modeling assumptions

The model relies on several simplifying assumptions:

- A fixed share α of trade occurs through long-term contracts.
- Demand and spot supply are linear.
- Demand and supply shocks are mean-zero and white noise.
- Contract prices are set using a simple forward-and-backward-looking rule.
- Buyers and sellers can be summarized by aggregate demand and supply schedules.

These assumptions make the model transparent. They also mean the paper abstracts from many features of the oil market, including storage, inventory speculation, detailed OPEC strategic behavior, political bargaining, and heterogeneous contract terms.

4.3 Why transitory shocks can have persistent effects

The model's persistence result does not require persistent physical shocks. It follows from the state variable p_{t-1}^C in the contract-price equation:

$$p_t^C = \phi p_{t-1}^C + \omega(\varepsilon_t^D - \varepsilon_t^S). \quad (30)$$

A one-time supply disruption raises p_t^C . Because next period's contract price depends on today's contract price, the effect carries forward. Persistence is therefore an institutional outcome, not necessarily a property of the original supply shock.

4.4 Why the spot price is not enough

A common intuition is that the spot price should be the marginal price and therefore should drive all price dynamics. Hubbard qualifies this view.

The relevant price for purchase decisions is:

$$P_t = \alpha P_t^C + (1 - \alpha) P_t^S, \quad (31)$$

not simply P_t^S . A buyer can use the spot market for short-term needs, but a contract purchase requires a longer-term commitment. Therefore spot-market information matters, but it does not mechanically determine the entire pricing system.

4.5 Why recursive regressions can fail

A regression of contract prices on lagged spot prices may fit one period well and fail in another because the underlying market structure changes. For example, if the spot-market share rises in the 1980s, the same physical shock should lead to a smaller spot-contract wedge than in the 1970s.

The model says the coefficients of

$$P_t^C = a_0 + a_1 P_{t-1}^C + a_2 P_{t-1}^S + e_t \quad (32)$$

are functions of deeper parameters such as α , f , g , π_1 , θ_1 , and γ . If those deeper parameters move, the reduced-form coefficients move too.

4.6 Policy implications

The paper discusses policy responses such as tariffs and public strategic stockpile releases. The key point is not that one specific policy is always best. The key point is that policy effects are regime-dependent.

Oil import tariffs. A tariff can reduce oil import demand or make demand more price responsive. In the model, a higher f lowers persistence:

$$\frac{\partial \phi}{\partial f} < 0. \quad (33)$$

So tariffs may reduce both the current impact and the persistence of shocks, at least in the simplified model.

Strategic stockpile releases. Stock releases can work like an increase in effective supply responsiveness. If they reduce the residual imbalance that the spot market must clear, they can dampen spot-price spikes. But their quantitative value depends on α and the contract-price adjustment rule.

Spot-price triggers. A policy rule that activates only when the spot price rises sharply can be misleading because the same spot-price movement has different implications under different contract regimes.

4.7 Limitations

The limitations are important:

- The contract share α is taken as exogenous, even though it likely responds to volatility, risk aversion, and transaction costs.
- The paper does not derive contract choice from buyer and seller optimization.
- The empirical exercise is simple and not a full structural estimate.
- The model abstracts from storage and inventory behavior, which are central in many commodity markets.
- The oil market has geopolitical features that are hard to capture in a compact price-adjustment model.

Bottom line. The paper's contribution is conceptual clarity about two-price dynamics, not precise numerical forecasts of oil prices.

5 Tables and figures

5.1 Table I: Crude oil spot and contract prices, 1973–1984

Read. Table I reports quarterly spot and contract prices for Mideast Light-34 crude oil. Selected observations show the scale of the spot-contract divergence:

Period	Spot price	Contract price
1973:1	2.08	2.10
1974:1	13.00	8.65
1979:4	38.17	22.84
1981:4	33.68	34.10
1984:4	27.81	28.74

Economic message. The spot market reacts sharply during disruptions. In 1974 and 1979, spot prices moved far above contract prices. But in the early 1980s, spot prices were often below contract prices. This two-sided wedge supports the idea that spot prices absorb short-term imbalances while contract prices adjust more slowly.

5.2 Figures I–IV: Dynamic response to a negative supply shock

Read. The paper's figures show the responses of spot and contract prices to a transitory negative supply shock under two persistence values, ϕ_0 and ϕ_1 , with $\phi_1 < \phi_0$.

RESPONSES OF OIL PRICES TO TRANSITORY NEGATIVE SUPPLY SHOCKS

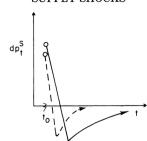


FIGURE I

Effect of a Transitory Shock on the Spot Price

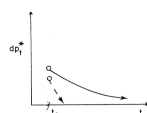


FIGURE II

Effect of a Transitory Shock on the Contract Price

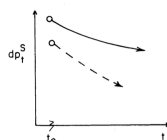


FIGURE III

Cumulative Effect of a Transitory Shock on the Spot Price

(continued on next)

RESPONSES OF OIL PRICES TO TRANSITORY NEGATIVE SUPPLY SHOCKS (CONTINUED)

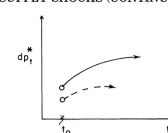


FIGURE IV

Cumulative Effect of a Transitory Shock on the Contract Price

- Figure I: the spot price jumps immediately.
- Figure II: the contract price rises more smoothly.
- Figure III: the cumulative spot-price effect depends on persistence.
- Figure IV: the cumulative contract-price effect is much larger when persistence is high.

Economic message. A shock can be physically temporary but economically persistent. Higher persistence means contract prices remain elevated for longer, which changes future spot-price clearing even after the original disruption is gone.

5.3 Table II: Instability of the spot-contract regression

Read. Table II estimates:

$$P_t^C = a_0 + a_1 P_{t-1}^C + a_2 P_{t-1}^S + e_t. \quad (34)$$

The estimates are:

Sample	a_0	a_1	a_2	R^2
1974:1–1984:12	0.333	0.906	0.081	0.975
1974:1–1979:12	2.767	0.576	0.187	0.937
1980:1–1984:12	0.916	0.897	0.072	0.935

Economic message. The coefficient on lagged contract prices is much lower before 1980 than after 1980, while the coefficient on lagged spot prices is larger before 1980. The relationship is not stable across regimes. This supports Hubbard’s warning that spot-contract regressions should not be treated as invariant policy models.

5.4 Connection between Table I and Table II

Table I shows large spot-contract wedges during turbulent periods. Table II shows that the statistical relation between those two prices shifts over time. Together they reinforce the same point: the oil market is not a simple one-price market with a stable adjustment equation.

6 Short summary

Hubbard studies how oil supply disruptions affect prices when the market has both spot transactions and long-term contracts. The paper’s central idea is that shocks are absorbed initially by the spot market, but contract prices adjust gradually using past prices, expected future prices, and demand-pressure information.

The model delivers a clean result: even white-noise supply shocks can have persistent effects on contract prices. The persistence comes from the contract-price adjustment rule, not necessarily from persistence in the physical shock itself.

The relevant price for demand is not only the spot price:

$$P_t = \alpha P_t^C + (1 - \alpha) P_t^S. \quad (35)$$

This weighted average makes the spot and contract markets jointly determined. A higher contract share makes the spot market smaller and increases the volatility and persistence effects of shocks.

The paper also explains why simple empirical models can be unstable. If the fraction of spot trades rises, or if contracts become shorter and more flexible, the relation between spot and contract prices changes. Therefore regressions that worked during the 1970s may fail in the 1980s.

7 Conclusion

7.1 Core conclusion

The world oil market's response to supply shocks depends on its pricing regime. When spot and contract prices coexist, a temporary disruption can create persistent price effects because contract prices adjust gradually.

7.2 Economic intuition

A small spot market must absorb short-term imbalances. This creates sharp spot-price movements during disruptions. But contract prices are updated more slowly, and once they move, they become a state variable for future price setting. The result is a dynamic system in which spot prices jump, contract prices drift, and the two prices interact over time.

7.3 Main theoretical contribution

The paper integrates short-run price adjustment and long-run contracting in one simple model. It shows that persistence can arise from price-setting institutions rather than from persistent shocks.

7.4 Main empirical lesson

The empirical relation between spot and contract prices is not structural. It changes when the market's contracting structure changes. This makes policy evaluation using fixed reduced-form price equations risky.

7.5 Takeaway

Hubbard's paper is an early and useful framework for thinking about commodity-market price rigidity. It shows that the impact of a supply shock is not determined only by the size of the shock, but also by the institutional architecture of market transactions.